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Indian Institute of Science Education and Research Mohali

CHM203 v.2 - Chemical kinetics [Dr. Jino George]

Midsem-2 Examination, March 3, 2026

Total marks: 20; Time: 9:30 am – 10:30 am

X 1. For a first-order parallel reaction, the Arrhenius factors for the formation of two products are 10^{10} and 10^8 sec^{-1} , and their energy of activation are 150 and 75 kJ mol^{-1} , respectively. At what temperature will the two products be formed at the same rate? (5 Marks)

X 2. Consider the decomposition reaction of HI. The collision diameter of HI = $3.5 \text{ \AA}$. Use collision theory to calculate the "steric factor P" at 500K, given that the measured pre-exponential factor is $6.7 \times 10^{12} \text{ M}^{-1}\text{sec}^{-1}$. (5 Marks)
[Hint: $\mu_{\text{HI}} = 1.65 \times 10^{-27} \text{ kg}$]

✓ 3. The pre-exponential factor for a bimolecular gas reaction occurring at 273°C is $6.0 \times 10^{10} \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$. Calculate the entropy of activation. (5 Marks)
[Hint: please use Arrhenius and collision theories]

X 4. Draw a hypothetical contour potential energy surface map for the following reaction:



(c) Represent the first-order saddle point.

(d) Draw the 2-D reaction coordinate of the same system as a separate plot and represent the activation energy. (2+1+2 Marks)

Important Formulae and universal constants:

1. $h = 6.626 \times 10^{-34} \text{ J s}$

2. $k_B = 1.38 \times 10^{-23} \text{ J/K}$

3. $c = 2.998 \times 10^8 \text{ m/s}$

4. $N_A = 6.023 \times 10^{23} \text{ /mol}$